

Supplementary materials: Prior-aware calibration of OpenSpliceAI-style splice-site prediction

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Supplementary methods

Target population and cache reconstruction

The schema-v2 evaluation included every test sequence and excluded all-zero padding before non-splice sampling. Every valid acceptor and donor position was retained. Valid non-splice positions were represented by a fixed reservoir sample and assigned a reconstruction weight equal to the number of valid non-splice positions divided by the number sampled. This reconstructs the population of valid positions in the GRCh38/MANE evaluation records; it does not represent every intergenic base in complete chromosomes.

Paired bootstrap design

Within each trained model, 200 bootstrap replicates were generated by stratified position-level resampling within the three true classes. The same resampled indices were applied to all calibration methods, so method contrasts were paired. Percentile intervals quantify uncertainty over the evaluated positions conditional on the fitted checkpoint and calibration parameters. They do not estimate variation across training seeds, genes, transcripts, or chromosomes.

Supplementary results

Table S1. Paired comparison of the two prior-aware calibration procedures

The contrast is OpenSpliceAI-style full-validation vector scaling relative to target-weighted true-logit vector scaling. Positive improvement favors the OpenSpliceAI-style procedure for every row; negative improvement favors target-weighted vector scaling. ECE, NLL, and Brier are lower-is-better, whereas AUPRC is higher-is-better.

Model	Metric	Improvement	Paired 95% interval	Replicates
Seed 11, epoch 11	Weighted multiclass ECE	2.576e-06	[-8.355e-06, 1.394e-05]	200
Seed 11, epoch 11	Weighted multiclass NLL	4.712e-08	[-3.782e-06, 3.899e-06]	200
Seed 11, epoch 11	Weighted multiclass Brier	4.038e-07	[-2.118e-06, 3.263e-06]	200
Seed 11, epoch 11	Target-weighted acceptor AUPRC	5.657e-05	[-1.831e-05, 1.274e-04]	200
Seed 11, epoch 11	Target-weighted donor AUPRC	2.696e-04	[-4.699e-04, 1.148e-03]	200
Seed 23, epoch 14	Weighted multiclass ECE	2.056e-06	[-7.272e-06, 1.198e-05]	200
Seed 23, epoch 14	Weighted multiclass NLL	-7.959e-07	[-5.815e-06, 5.014e-06]	200

Model	Metric	Improvement	Paired 95% interval	Replicates
Seed 23, epoch 14	Weighted multiclass Brier	-9.506e-07	[-4.573e-06, 2.675e-06]	200
Seed 23, epoch 14	Target-weighted acceptor AUPRC	6.072e-04	[-1.324e-04, 1.580e-03]	200
Seed 23, epoch 14	Target-weighted donor AUPRC	-2.224e-04	[-8.666e-04, 3.622e-04]	200

None of the ten paired intervals excludes zero. These results do not establish superiority or equivalence of the two procedures.

Table S2. Evaluation population and analysis provenance

Item	Value
Test sequences	84,258
Raw fixed-length test positions	421,290,000
Excluded all-zero padding positions	14,011,378
Valid test positions	407,278,622
Valid non-splice positions	407,170,634
Valid acceptor positions	53,994
Valid donor positions	53,994
Sampled valid non-splice positions per cache	500,000
Test non-splice reconstruction weight	814.341268
Validation non-splice reconstruction weight	177.980664
Reliability bins	15 equal-width bins
Bootstrap replicates	200
Bootstrap seed	17
Bootstrap unit	Position
Resampling	Stratified by true class and paired across methods

The authoritative point-estimate file hashes were:

- Seed 11, epoch 11: a35b673ba71b60e550ca288ca915740b1591854553b777006902c76f54649882
- Seed 23, epoch 14: f2cb2e59ae9a514e9ebec8c4aa2f6f975bc4e5679d65f7464e12e33e6f7043cc

The corresponding schema-v2 cache hashes were:

- Seed 11, epoch 11: 31387e49d9f3de563ab6084c13db07b4915346265a22fbbe5d3b2830c7be4b1a
- Seed 23, epoch 14: 7acf7e9e98a534c5c368b39a5ffca95bdd73004f098459fc51b88cf72f677e5e

Reliability-bin support

Reliability curves use 15 equal-width bins. Curves connect only nonempty bins, so visual line length and jaggedness must not be interpreted as equal evidential support. The binary acceptor and donor panels contain bins with very small reconstructed target-population mass, particularly at higher probabilities. Their raw gap can be visually large while their contribution to target-weighted ECE remains small.

For each seed, the machine-readable reliability data retain the method, reliability type, class, bin limits, raw count, reconstructed target weight, target-weight fraction, mean predicted probability, observed frequency or accuracy, absolute gap, and ECE contribution. These records belong in Supplementary Data rather than a printed table because five methods, three reliability views, and 15 bins produce hundreds of rows per model.

Developmental flank-80 workflow

The seed-11/epoch-2 flank-80 configuration was used during development to establish memory-safe model training and true-logit extraction. It was later re-extracted with schema-v2 rules. The audit confirmed complete coverage of 18,486 validation sequences and 84,258 test sequences, exclusion of 3,413,758 validation and 14,011,378 test padding positions, and the same valid test label census used for flank-400.

The earlier flank-80 calibration results were generated before schema v2, and the full five-method calibration plus paired-bootstrap analysis was not repeated on the corrected caches. Those point estimates and figures are therefore not included as final quantitative evidence. The study does not estimate the causal effect of context length.

Machine-readable supplementary data

- **Supplementary Data S1:** full 13-metric point estimates for five methods, two checkpoints, and both sampled-cache and target-weighted AUPRC definitions.
- **Supplementary Data S2:** changes relative to each checkpoint’s uncalibrated result, including absolute and percentage changes with metric direction.
- **Supplementary Data S3:** paired prior-aware contrasts underlying Table S1.
- **Supplementary Data S4:** reliability-bin tables, reliability summaries, and generation metadata for both checkpoints.
- **Supplementary Data S5:** cross-seed metadata and SHA-256 provenance.

Sampled-cache AUPRC values in Supplementary Data S1 describe the splice-enriched cache and must not be substituted for the target-weighted AUPRC values in the main text.